

Assignment - 5

Name: Sciddhanto Sinha

Class: CSE, SY-2(B)

Roll No.:2213111

Enrollment No.: MITU21BTCS0553

Aim : (Lists)

a. Write a Python program to find a list of Strings with exactly two occurrences of “MIT” and at least one occurrence of “ADT”.

Input: [“MIT”, “SOE”, “MIT”, “ADTU”, “ADT”, “Loni”, “Design”, “Food”, “Technology”]

b. Extend the program for the following integer inputs.

Input: [100, 35, 23, 100, 45, 89, 90] //Exactly 2 occurrences of 100 and at least 2 occurrences of 90

Input: [90, 110, 130, 150, 170, 200, 200] //At Least 2 occurrences of 200 and exactly one occurrence of 130

c. Write a Python program that accepts a list of integers and checks the length and the

third element. Return true if the length of the list is more than 10 and the third element occurs twice in the said list.

d. Sort the list in ascending order.

Theory : In this work we will search elements from strings and list of integers using nested **if** Inside a **for** loop using a variable count to initialize the loop and incrementing that variable if the required string or integer is in the list

Program :

```
"""
Name :Sciddhanto
Class : SY-22-B
Roll No. : 2213825
"""

#Task A
print("=" * 150)
print("\nTask A\n")
str_lst = ["MIT", "SOE", "MIT" , "ADTU", "ADT", "Loni", "Design", "Food",
"Technology"]
print("The list of string is  :: ",str_lst)
check_str = str(input("Enter string element to check :: "))

count = 0
for x in str_lst:
    if (x == check_str):
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        count = count + 1
print("{} has occurred {} times in the given list ".format(check_str,
count))

#Task B
print("=" * 150)
print("\nTask B\n")
int_lst = [100, 35, 23, 100, 45, 89, 90]
print("The list of integer is  :: ",int_lst)
check_int = int(input("Enter integer element to check :: "))

count_int = 0
for y in int_lst:
    if (y == check_int):
        count_int = count_int + 1
print("{} has occurred {} times in the given list ".format(check_int,
count_int))

#Task C
print("=" * 150)
print("\nTask C\n")
def test(lst):
    # return len(lst) >= 10 and lst.count(lst[2]) == 2
    if len(lst) <= 10:
        print("List length is not more than 10")
    elif lst.count(lst[2]) >= 2:
        print("List has length more than 10 and {} occurs 2 or more
times".format(lst[2]))
    else:
        print("ERROR!!")

lst = []
n = int(input("Number of elements in array :: "))
lst = []
for i in range(0,n):
    enter = int(input("Enter List Contents :: "))
    lst.append(enter)

```

```

print("Original list :: ")
print(lst)
print("Check whether the length of the said list is 10 and third element
occurs twice in the said list. :: ")
print(test(lst))

#Task D
print("=" * 150)
print("\nTask D\n")
int_lst.sort()
print("The sorted list of integers is :: ",int_lst)
print("=" * 150)

```

Output :

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=====
Task A

The list of string is :: ['MIT', 'SOE', 'MIT', 'ADTU', 'ADT', 'Loni',
'Design', 'Food', 'Technology']
Enter string element to check :: MIT
MIT has occurred 2 times in the given list
=====

Task B

The list of integer is :: [100, 35, 23, 100, 45, 89, 90]
Enter integer element to check :: 35
35 has occurred 1 times in the given list
=====

Task C

Number of elements in array :: 11
Enter List Contents :: 1
Enter List Contents :: 1
Enter List Contents :: 1

```

```
Enter List Contents :: 2
Enter List Contents :: 3
Enter List Contents :: 4
Enter List Contents :: 5
Enter List Contents :: 6
Enter List Contents :: 7
Enter List Contents :: 8
Enter List Contents :: 9
Original list ::
[1, 1, 1, 2, 3, 4, 5, 6, 7, 8, 9]
Check whether the length of the said list is 10 and third element occurs
twice in the said list. ::
List has length more than 10 and 1 occurs 2 or more times
None
```

Task D

```
The sorted list of integers is :: [23, 35, 45, 89, 90, 100, 100]
```

Conclusion :

In this assignment we learnt how to search for the string and integers from a list of strings and integers as per our requirements and sorting those list in ascending order